SpaceX Falcon 9 landing prediction

Joao Trevisan | 23 October 2026

Question: Which Falcon 9 launch characteristics are associated with a successful first-stage landing, and how well can classifiers predict the outcome?

Public project repository: completed notebooks, Python scripts, data and results

https://github.com/joaoltrevisan/spacex-falcon9-capstone

Decision Tree: best five-fold CV accuracy 87.7%; 83.3% holdout accuracy. Full evidence is presented in the following slides.

Executive Summary

• API data collection; historical web scraping; feature wrangling; six EDA charts; ten SQL analyses; Folium mapping; interactive Plotly Dash; four tuned classifiers.

• 90-row modeling sample: 60/90 successful landings (66.7%). Success rates vary by site, orbit and year; categories and time periods are imbalanced.

• 10-query SQL task results are shown individually on slides 21-30. Folium and Dash evidence appears on slides 31-36.

• Decision Tree led five-fold CV (0.877) and holdout accuracy (0.833). Confusion matrix: TN=4, FP=2, FN=1, TP=11; precision=.846, recall=.917, F1=.880.

• Independent samples have different denominators: Dash 24/56=42.9%; modeling 60/90=66.7%. They must not be combined. A selected 2,000-6,000 kg Dash subset is 18/34=52.9%.

Introduction: context and questions

• Reusability of the Falcon 9 first stage is relevant to launch economics and recovery operations.

• Target: booster landing success (Class 1) versus other recorded landing outcomes (Class 0). Mission payload delivery success is a different outcome.

• Questions: how do landing outcomes vary with flight number, launch site, orbit and payload? What do the ten SQL tasks reveal? Which model performs best on a held-out sample?

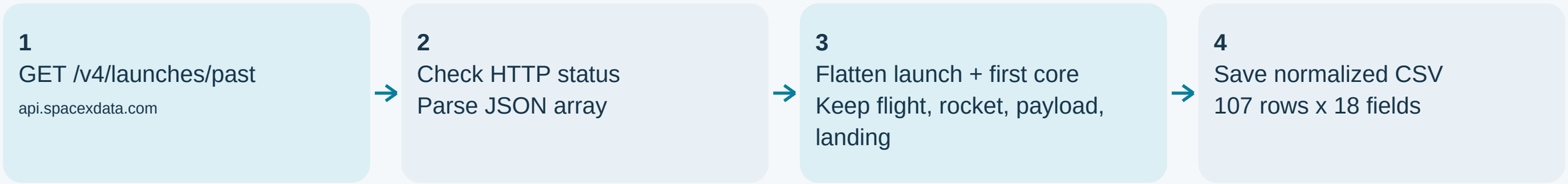
• Historical, observational analysis can identify associations; it cannot establish that a particular site, payload or orbit causes a successful landing.

• The final repository and each completed notebook are linked on the relevant methodology slides.

Workflow and datasets

- Archived IBM SpaceX API response: 107 launch records, normalized to 18 fields. The notebook attempts the documented live REST GET and transparently falls back to the IBM archive if unreachable.
- Wikipedia revision 987426997: 97 dated launch rows through 2020-11-05.
- IBM modeling table plus supplied 90 x 80 feature matrix: validated and used for six EDA plots and predictive models.
- SQL lab dataset: 101 records. Folium geography dataset and Dash dataset: 56 records each.
- Every result below names its sample. No unsupported join combines these incompatible samples.

Data collection: SpaceX REST API



Nested response fields include launch date, mission success, rocket and launchpad IDs, payload IDs, core reuse, landing attempt/type/result and failures. The notebook attempts the live request; when unreachable it uses the fixed IBM API response and reports that fallback.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/01_Data_Collection_API.ipynb

SpaceX API: request and output evidence

GET https://api.spacexdata.com/v4/launches/past

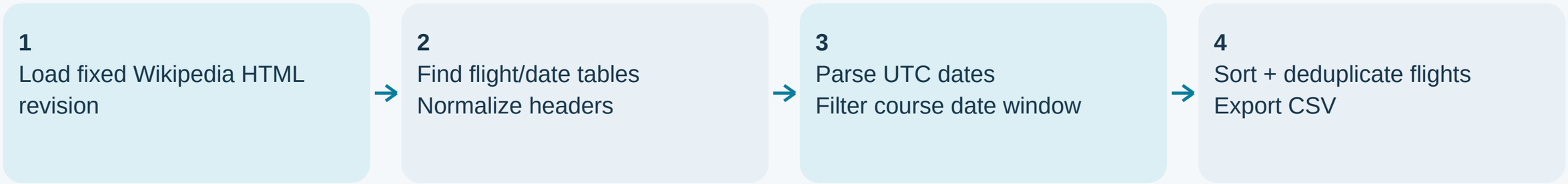
Python: requests.get(API_URL, timeout=5) -> inspect response.status_code -> raise_for_status() -> response.json() -> validate list -> normalize records -> export CSV.

Reproducible outcome: IBM archive contains 107 records and the normalized output has 107 rows x 18 fields. The code records whether the current run used live JSON or the IBM fallback.

The public SpaceX API endpoint is community-maintained. The saved course response makes the analysis reproducible when the live service is unreachable.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/01_Data_Collection_API.ipynb

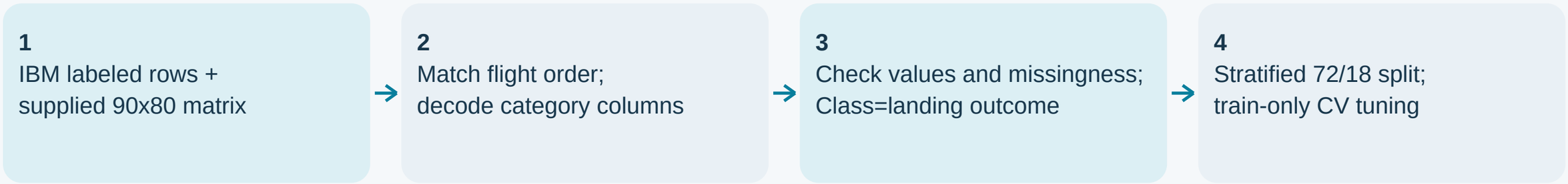
Data collection: historical web scraping



Revision 987426997 | Date window: 2010-06-04 through 2020-11-05 | Output: 97 unique dated launch rows. The fixed source revision prevents the live page from changing this historical sample.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/02_Data_Collection_Web_Scraping.ipynb

Data wrangling: validate features and target



All 90 flights and 80 feature values validated against the IBM source; one-hot Orbit, LaunchSite, LandingPad and Serial categories align. Class 1=success (60); Class 0=other outcome (30).

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/03_Data_Wrangling_and_Target.ipynb

EDA with visualization: six required charts

- Scatter plots: Flight Number vs Launch Site; Payload vs Launch Site; Flight Number vs Orbit; Payload vs Orbit. Color encodes Class 0/1.
- Bar chart: success rate by orbit, with the count for each orbit to show small denominators.
- Line chart: annual average landing success rate, annotated with yearly counts.
- Use: examine launch sequence, site/orbit mix and payload patterns. These unadjusted comparisons are descriptive, not causal.
- Six separate result slides with interpretation follow (slides 15-20).

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/04_EDA_and_Feature_Engineering.ipynb

EDA with SQL: query method

1
Load 101-row Spacex.csv

→

2
Trim string whitespace; SQLite table

→

3
SELECT, LIKE, aggregates, subquery, DENSE_RANK

→

4
Return each query + results + explanation

Landing_Outcome (booster recovery) remains distinct from Mission_Outcome (payload delivery). Slide results 21-30 use this lab sample only.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

Interactive mapping with Folium

• Circle and marker for each launch site; tooltip/popup shows the site and launches.

• MarkerCluster preserves 56 individual launch records at overlapping Florida coordinates; green=successful landing, red=unsuccessful landing.

• Polyline between CCAFS LC-40 and selected coastline/road points displays approximate straight-line distances. Haversine computes distance.

• MousePosition helps read coordinates on visible map tiles. Three result screenshots are provided on slides 31-33.

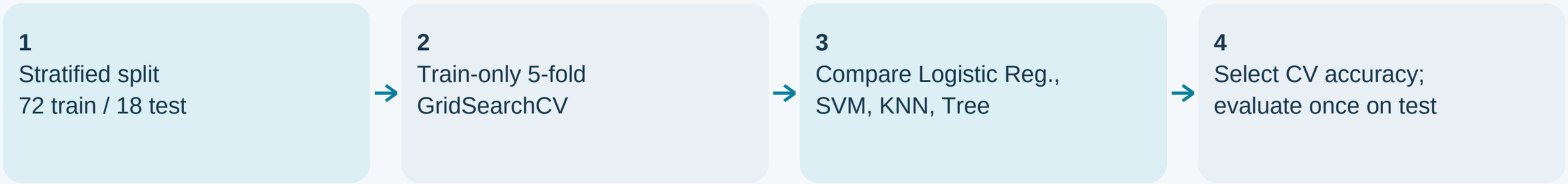
https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

Dashboard with Plotly Dash

- Pie chart counts successful launches across all sites (24 successes); selecting a site changes the pie to its success/failure ratio.
- Scatter plot shows Payload Mass vs binary landing outcome; marker color encodes booster version; hover shows flight and site.
- Launch-site dropdown filters the selected site. Payload RangeSlider filters scatter records and updates visible sample size.
- The pie denominator and scatter range are stated separately. Result screenshots on slides 34-36.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/src/dash_app.py

Predictive analysis: classifier development



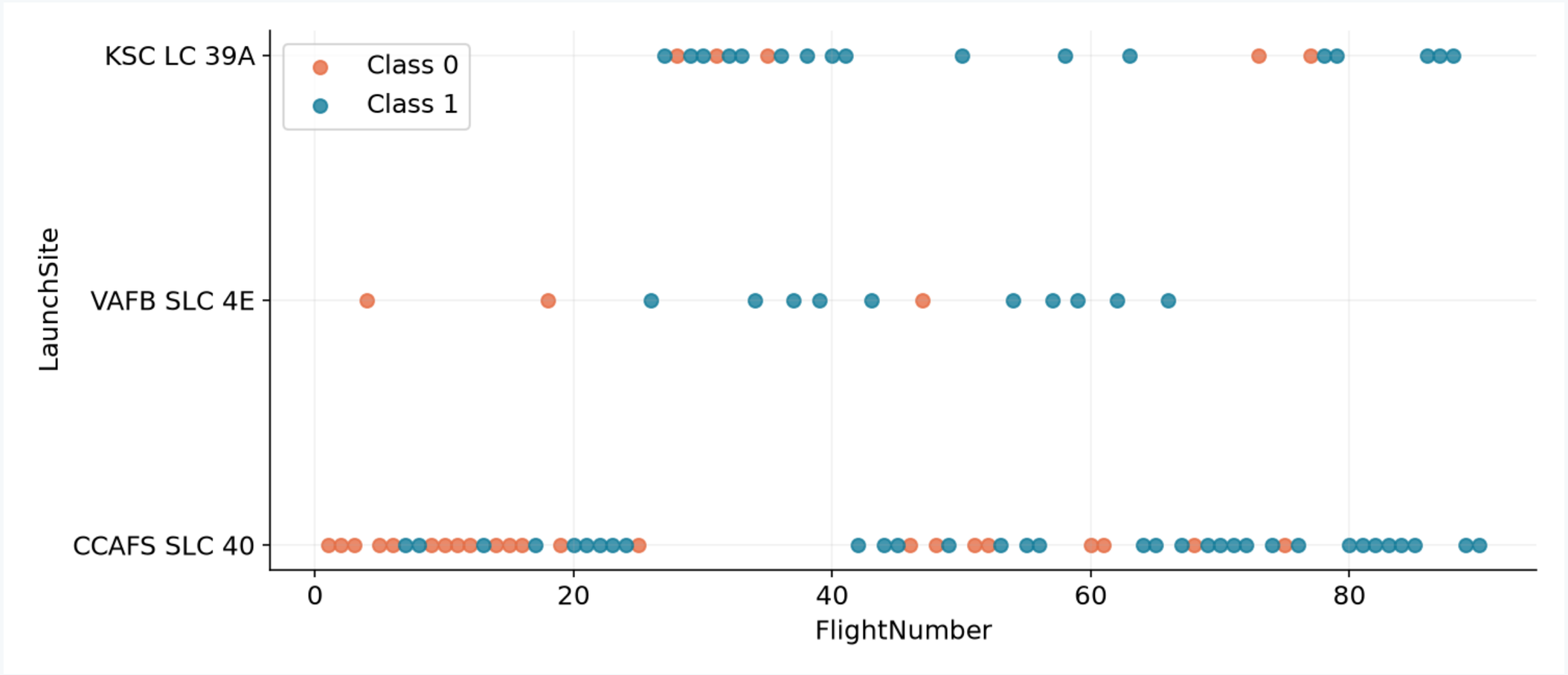
Scaling is inside the Logistic Regression/SVM/KNN pipelines. Tree grid tunes Gini/depth/leaf. Test metrics do not choose hyperparameters. Full parameter table and scores appear on slide 37.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/07_Modeling.ipynb

EDA and predictive results

- Six visualization results from the validated 90-row labeled sample: site, orbit, payload, flight sequence and annual trends.
- Ten separate SQL query outcomes on the 101-row laboratory dataset.
- Interactive Folium maps show locations, landing results and approximate proximity.
- Dash shows success counts/ratios and payload-range behavior from its own 56-record sample.
- Four classifier metrics and the best model confusion matrix use the 18-record holdout.

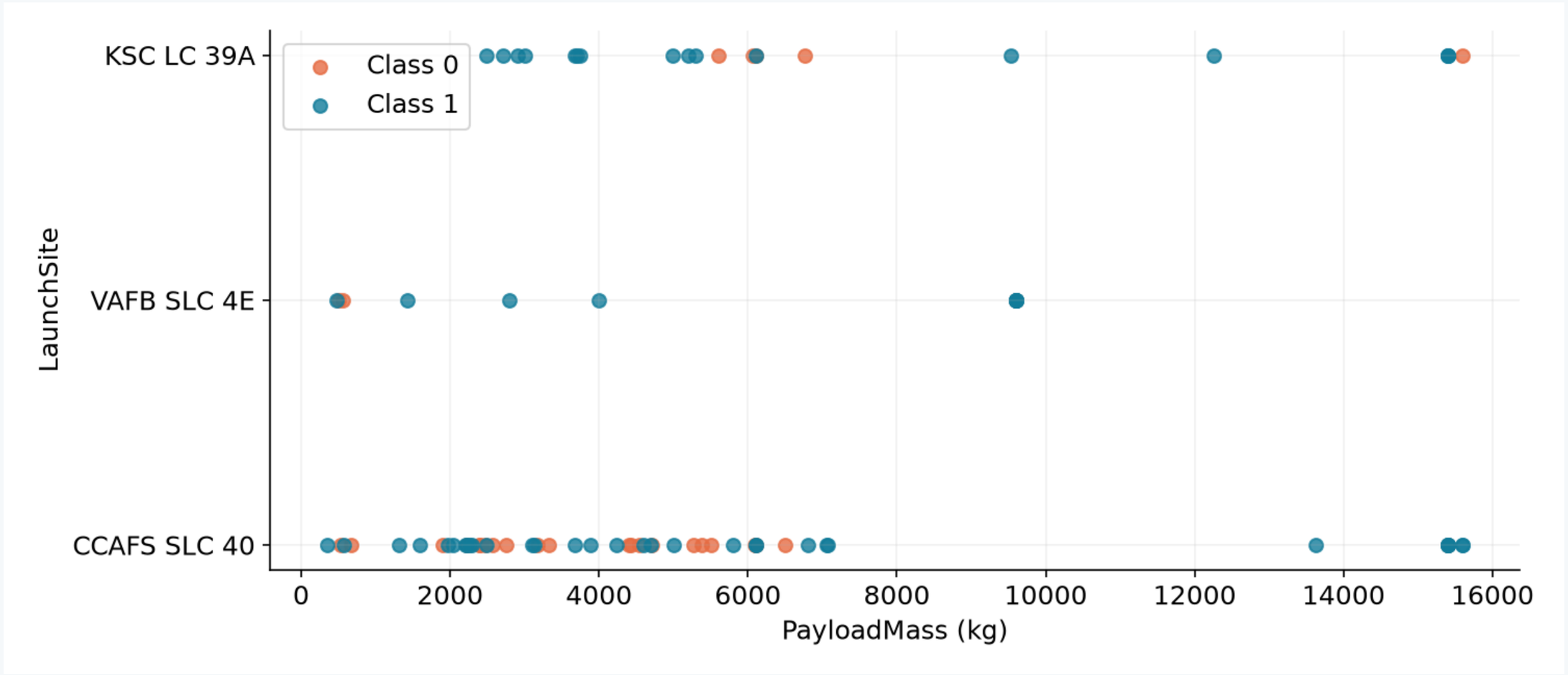
Flight Number vs Launch Site



Later flights show more successes across sites. KSC has 17/22 (77.3%), VAFB 10/13 (76.9%), CCAFS SLC 40 33/55 (60.0%). Site results reflect unequal sample size and mission mix.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/04_EDA_and_Feature_Engineering.ipynb

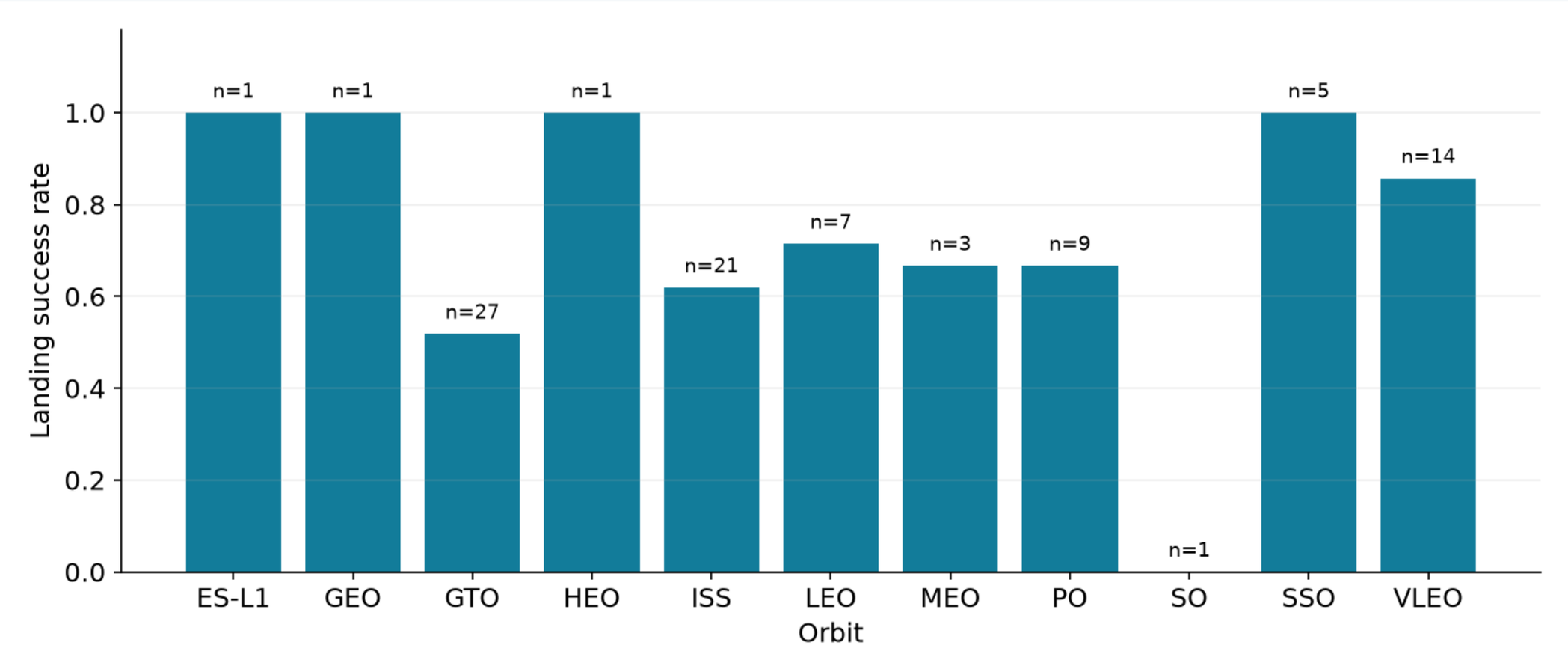
Payload vs Launch Site



Both outcomes appear across payload values at each site. Payload distributions differ by site; mission mix limits a direct site comparison.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/04_EDA_and_Feature_Engineering.ipynb

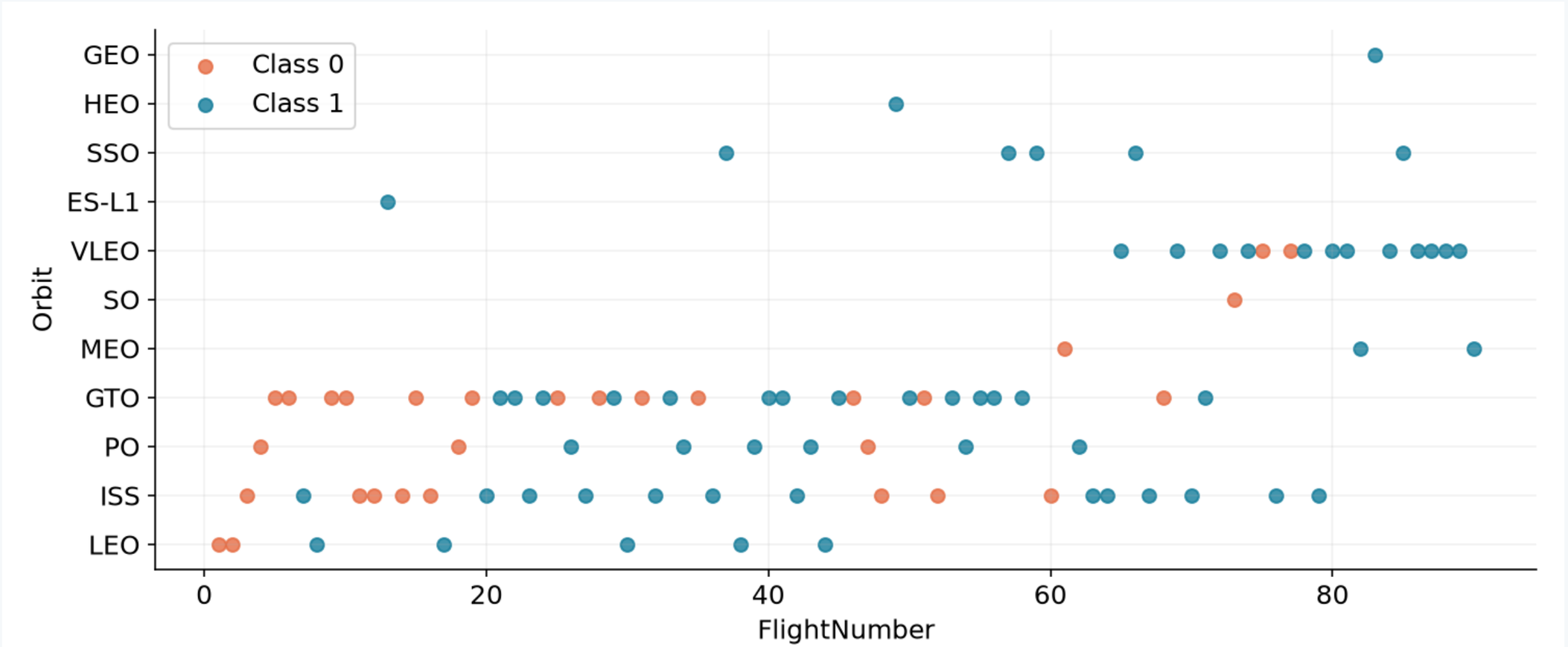
Success Rate vs Orbit Type



SSO 5/5, VLEO 12/14, GTO 14/27. Several 100% orbits have one flight only. Small denominators make those rates unstable.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/04_EDA_and_Feature_Engineering.ipynb

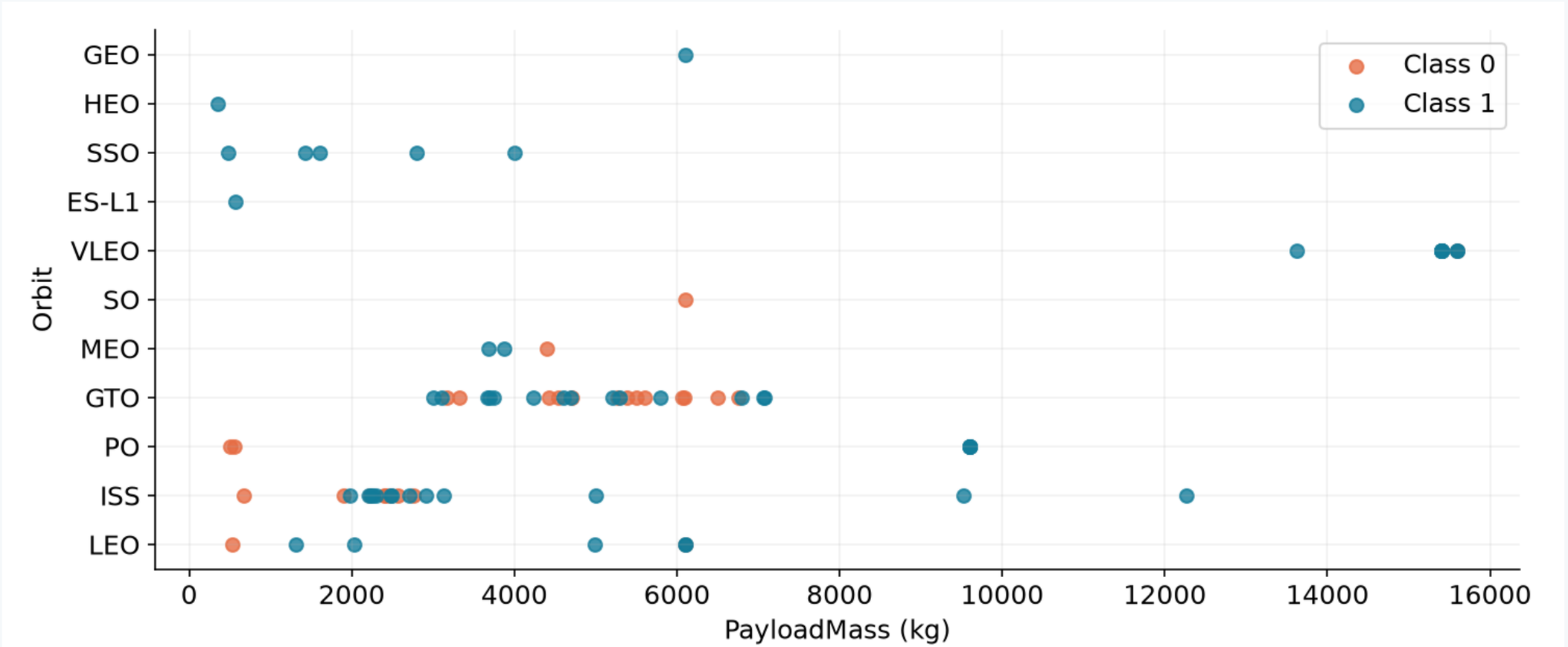
Flight Number vs Orbit Type



Orbit types are unevenly distributed over mission sequence. Experience and hardware changes can confound the apparent relationship.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/04_EDA_and_Feature_Engineering.ipynb

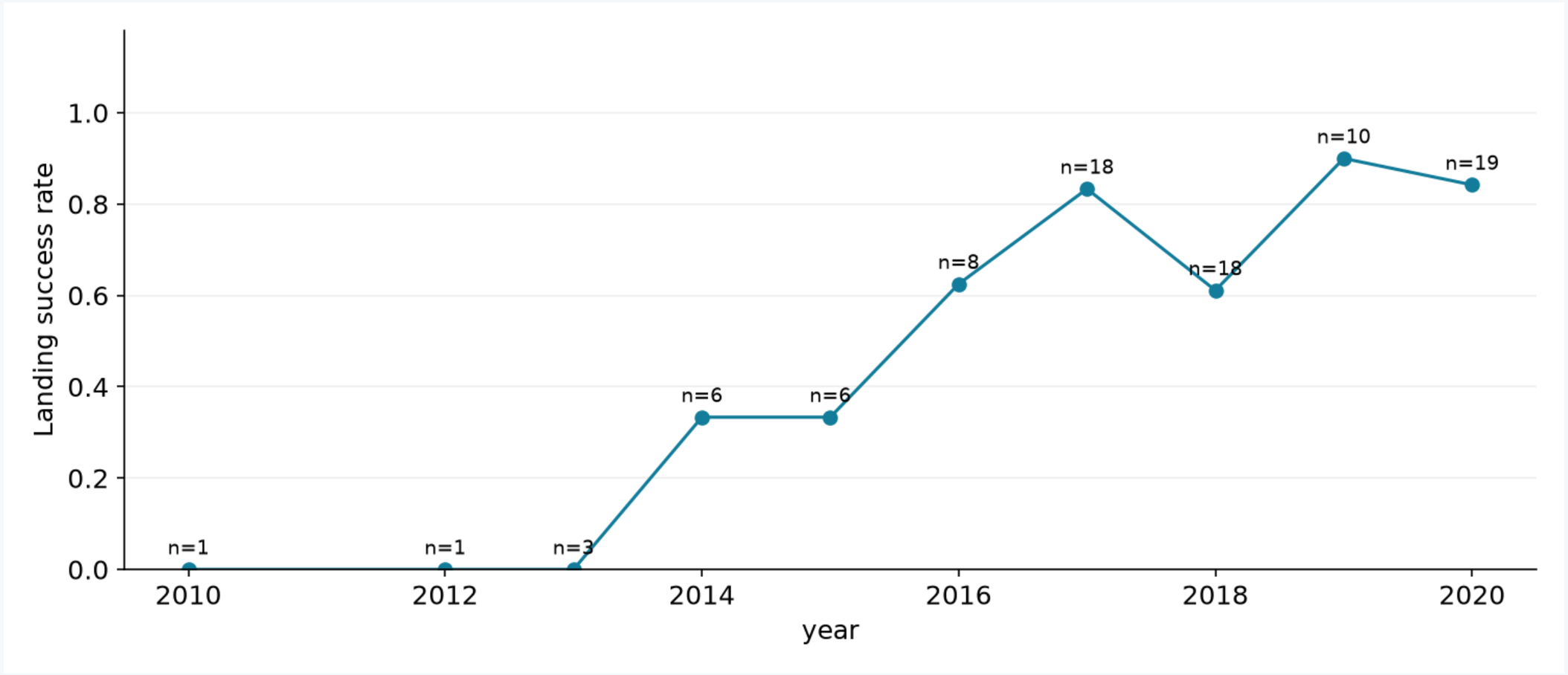
Payload vs Orbit Type



Payload mass varies by orbit and mission requirements. The graph does not establish that higher payload causes successful landing.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/04_EDA_and_Feature_Engineering.ipynb

Launch Success Yearly Trend



Annual rates generally rise over time but are not monotonic: 2017=15/18, 2018=11/18, 2019=9/10, 2020=16/19. Early years are sparse.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/04_EDA_and_Feature_Engineering.ipynb

All Launch Site Names

Query

SELECT DISTINCT Launch_Site FROM launches ORDER BY Launch_Site

Launch Site
CCAFS LC-40
CCAFS SLC-40
KSC LC-39A
VAFB SLC-4E

Result and interpretation: Four distinct location labels are retained as recorded in the dataset.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

Launch Site Names Begin with CCA

Query

SELECT Date, Booster_Version, Launch_Site, PAYLOAD_MASS__KG_ FROM launches WHERE Launch_Site LIKE 'CCA%' ORDER BY Date LIMIT 5

Date	Booster Version	Launch Site	PAYLOAD MASS KG
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	0
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	0
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	525
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	500
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	677

Result and interpretation: The five earliest matching records are displayed in chronological order.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

Total Payload Mass from NASA CRS

Query

SELECT Customer, SUM(PAYLOAD_MASS__KG_) AS payload_kg FROM launches WHERE Customer = 'NASA (CRS)' GROUP BY Customer

Customer	payload kg
NASA (CRS)	45596

Result and interpretation: NASA (CRS) customers carry 45,596 kg total. A separate LIKE NASA query sums all NASA-named customers to 107,010 kg across 32 records.

NASA CRS exact category is the course question. The all-NASA result is supplementary and does not replace it.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

Average Payload Mass by F9 v1.1

Query

SELECT COUNT(*) AS launches, ROUND(AVG(PAYLOAD_MASS__KG_), 2) AS mean_payload_kg FROM launches WHERE Booster_Version LIKE 'F9 v1.1%'

launches	mean payload kg
15	2534.67

Result and interpretation: F9 v1.1 carries an average 2,534.67 kg across 15 launches.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

First Successful Ground Landing Date

Query

```
SELECT MIN(Date) AS first_successful_ground_landing FROM launches WHERE Landing_Outcome = 'Success (ground pad)'
```

first successful ground landing
2015-12-22

Result and interpretation: First ground-pad landing success: 2015-12-22.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

Successful Drone Ship Landing: payload 4,000-6,000 kg

Query

SELECT Date, Booster_Version, PAYLOAD_MASS__KG_ FROM launches WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000 ORDER BY Date

Date	Booster Version	PAYLOAD MASS KG
2016-05-06	F9 FT B1022	4696
2016-08-14	F9 FT B1026	4600
2017-03-30	F9 FT B1021.2	5300
2017-10-11	F9 FT B1031.2	5200

Result and interpretation: Four successful drone-ship landings satisfy the strict payload bounds.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

Total Successful and Failure Mission Outcomes

Query

SELECT Mission_Outcome, COUNT(*) AS missions FROM launches GROUP BY Mission_Outcome ORDER BY missions DESC

Mission Outcome	missions
Success	99
Success (payload status unclear)	1
Failure (in flight)	1

Result and interpretation: Mission outcomes concern payload delivery: 99 Success, 1 Success (payload status unclear), 1 Failure (in flight).

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

Boosters Carried Maximum Payload

Query

SELECT Booster_Version, PAYLOAD_MASS__KG_ FROM launches WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM launches) ORDER BY Date

Booster Version	PAYLOAD MASS KG
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

Result and interpretation: Maximum payload is 15,600 kg; 12 launches share the maximum.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

2015 Failed Drone Ship Landing Records

Query

```
SELECT Date, substr(Date,6,2) AS month, Booster_Version, Launch_Site, Landing_Outcome FROM launches WHERE Date LIKE '2015-%' AND Landing_Outcome = 'Failure (drone ship)'
ORDER BY Date
```

Date	month	Booster Version	Launch Site	Landing Outcome
2015-01-10	01	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
2015-04-14	04	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Result and interpretation: Two failures occurred Jan 10 and Apr 14; both launched from CCAFS LC-40.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

Rank Landing Outcomes: 2010-06-04 to 2017-03-20

Query

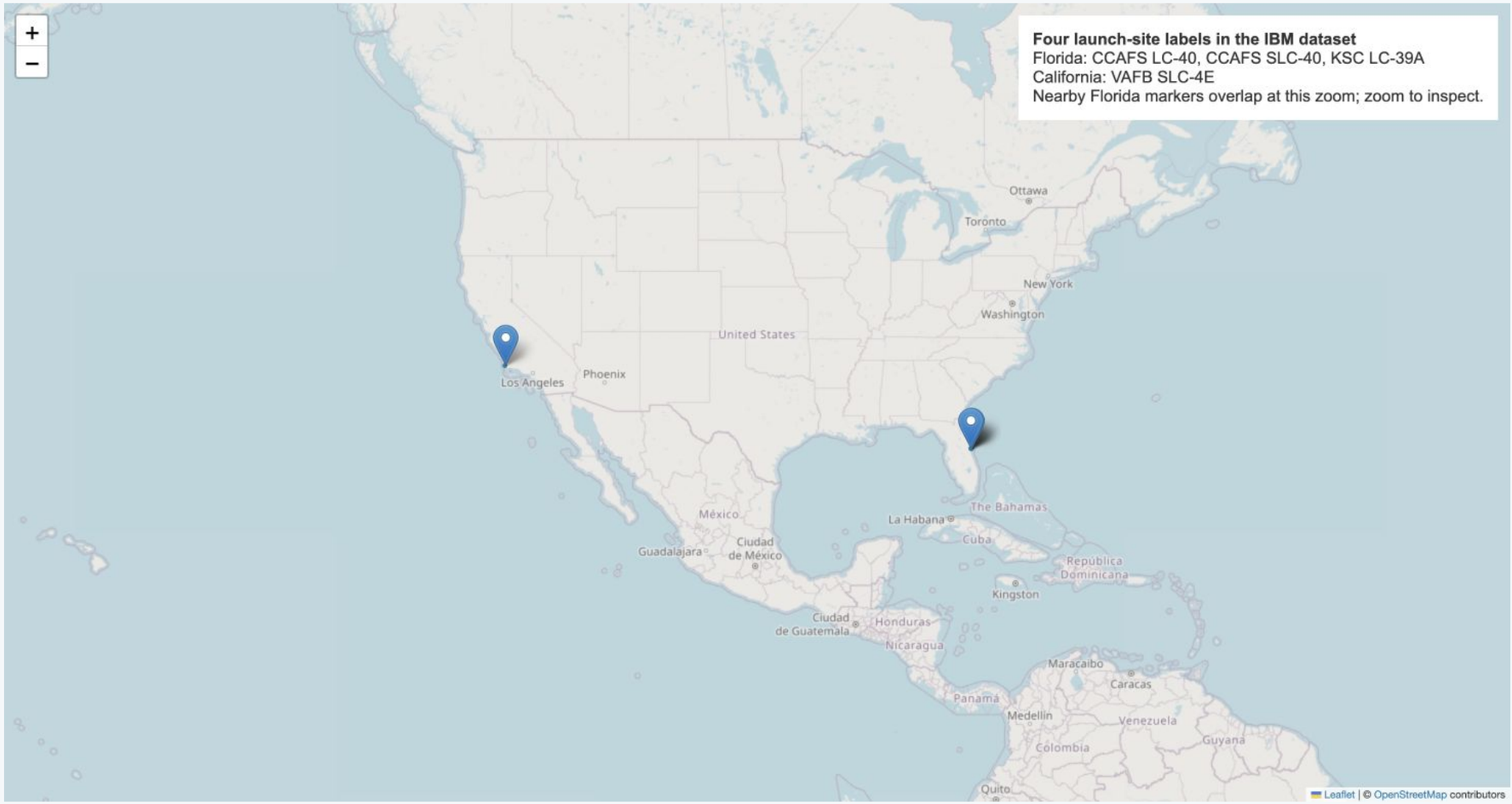
```
SELECT Landing_Outcome, COUNT(*) AS launches, DENSE_RANK() OVER (ORDER BY COUNT(*) DESC) AS rank FROM launches WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome ORDER BY launches DESC, Landing_Outcome
```

Landing Outcome	launches	rank
No attempt	10	1
Failure (drone ship)	5	2
Success (drone ship)	5	2
Controlled (ocean)	3	3
Success (ground pad)	3	3
Failure (parachute)	2	4
Uncontrolled (ocean)	2	4
Precluded (drone ship)	1	5

Result and interpretation: Dense ranking preserves ties. Date endpoints are inclusive.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

All launch sites on a global map



The four IBM dataset labels span Florida and California. Florida launch-pad markers overlap at national zoom; interactive zoom and marker popups distinguish them.

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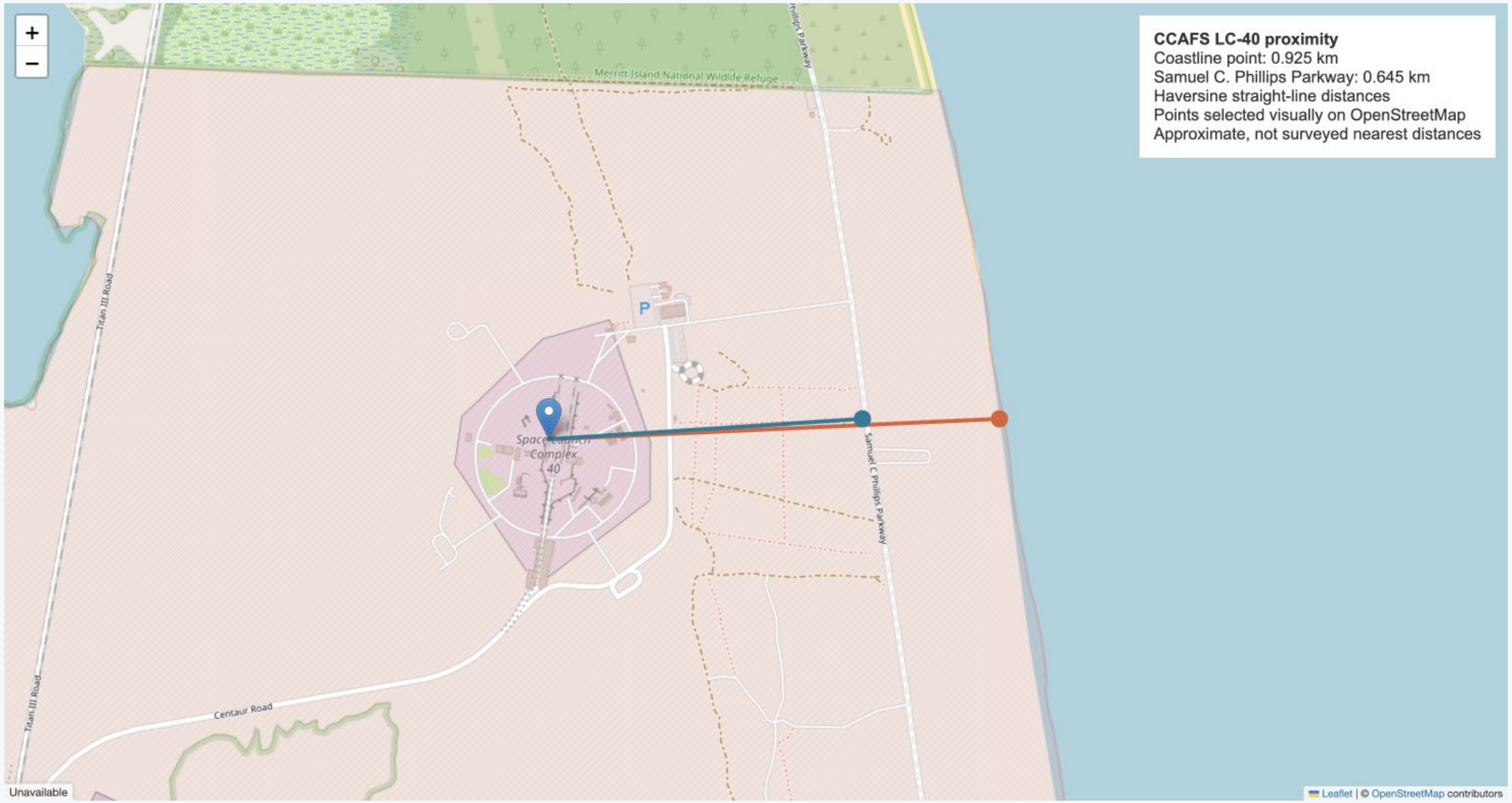
Color-coded launch outcomes on Folium



At KSC, the expanded cluster displays 13 individual launch records: 10 successful outcomes (green) and 3 failures (red).

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

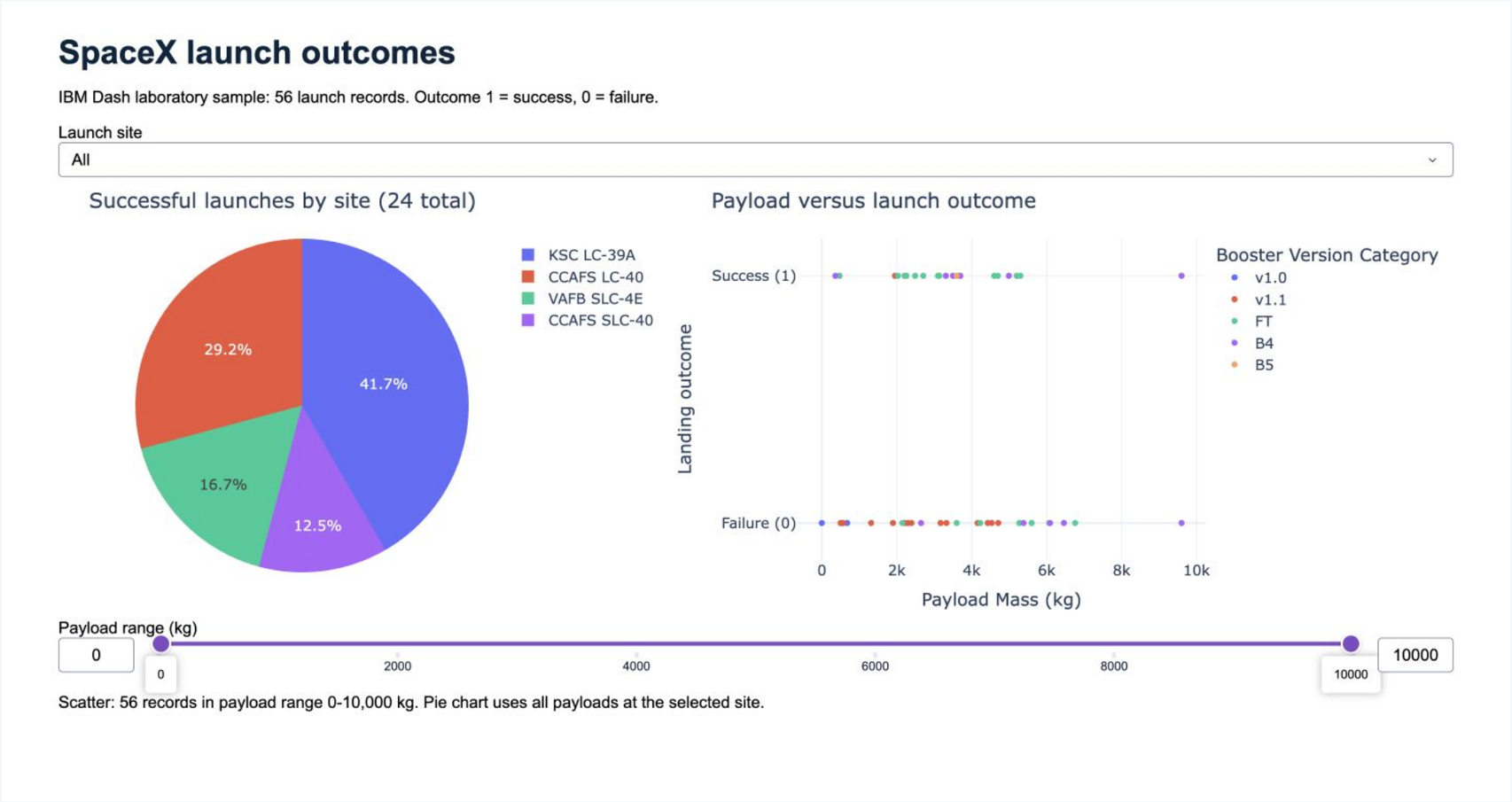
Selected launch-site proximity to coast and road



From CCAFS LC-40: 0.925 km to selected coastline point and 0.645 km to Samuel C. Phillips Parkway. Haversine straight-line distance; visually selected approximate points, not surveyed routes.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/05_SQL_EDA_and_Folium_Map.ipynb

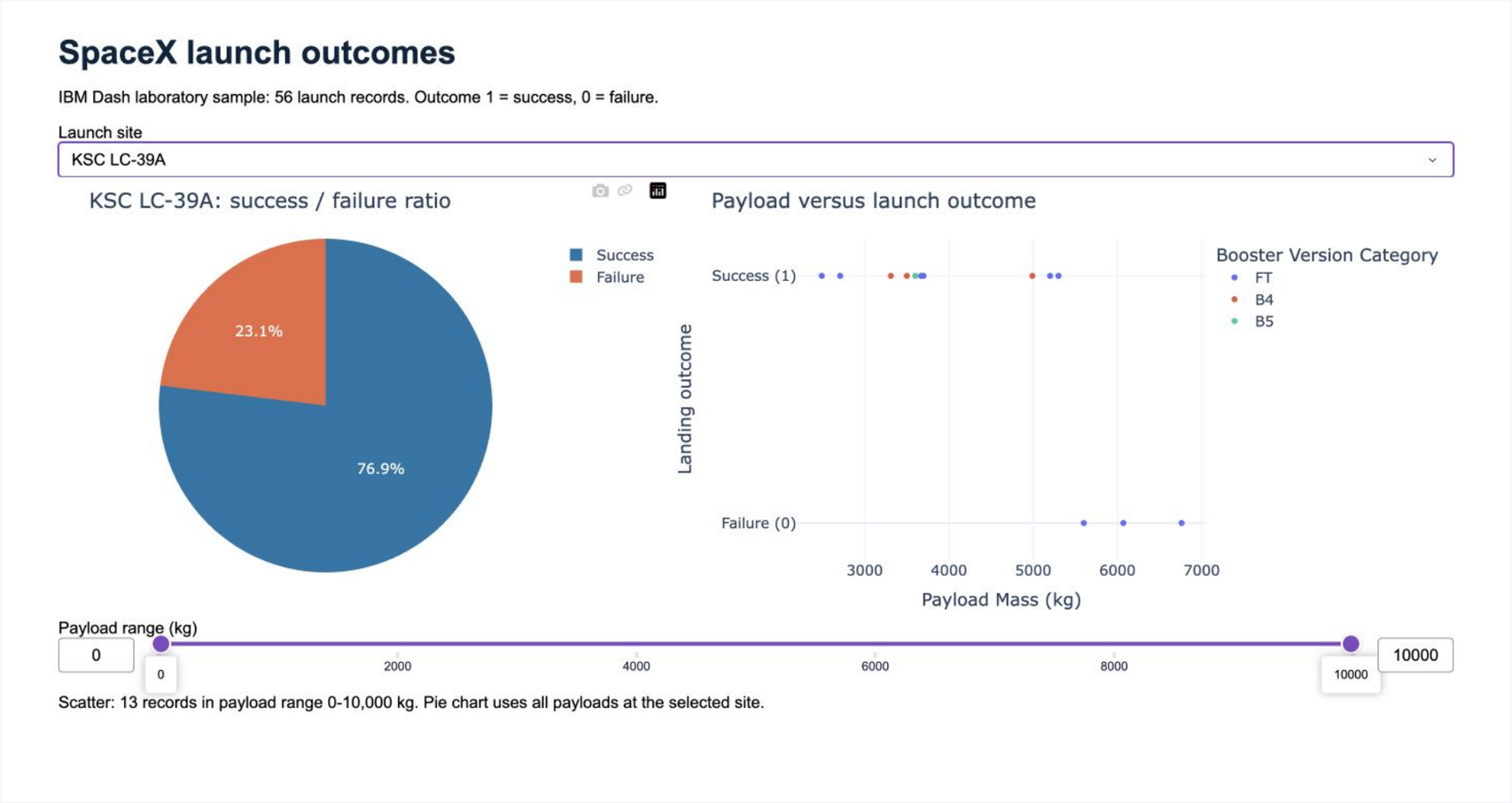
Successful launch count by site



All-site pie chart: 24 successful launches: KSC 10 (41.7%), CCAFS LC-40 7 (29.2%), VAFB 4 (16.7%), CCAFS SLC-40 3 (12.5%). Counts of success, not site-specific rates.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/06_Plotly_Dash_Dashboard.ipynb

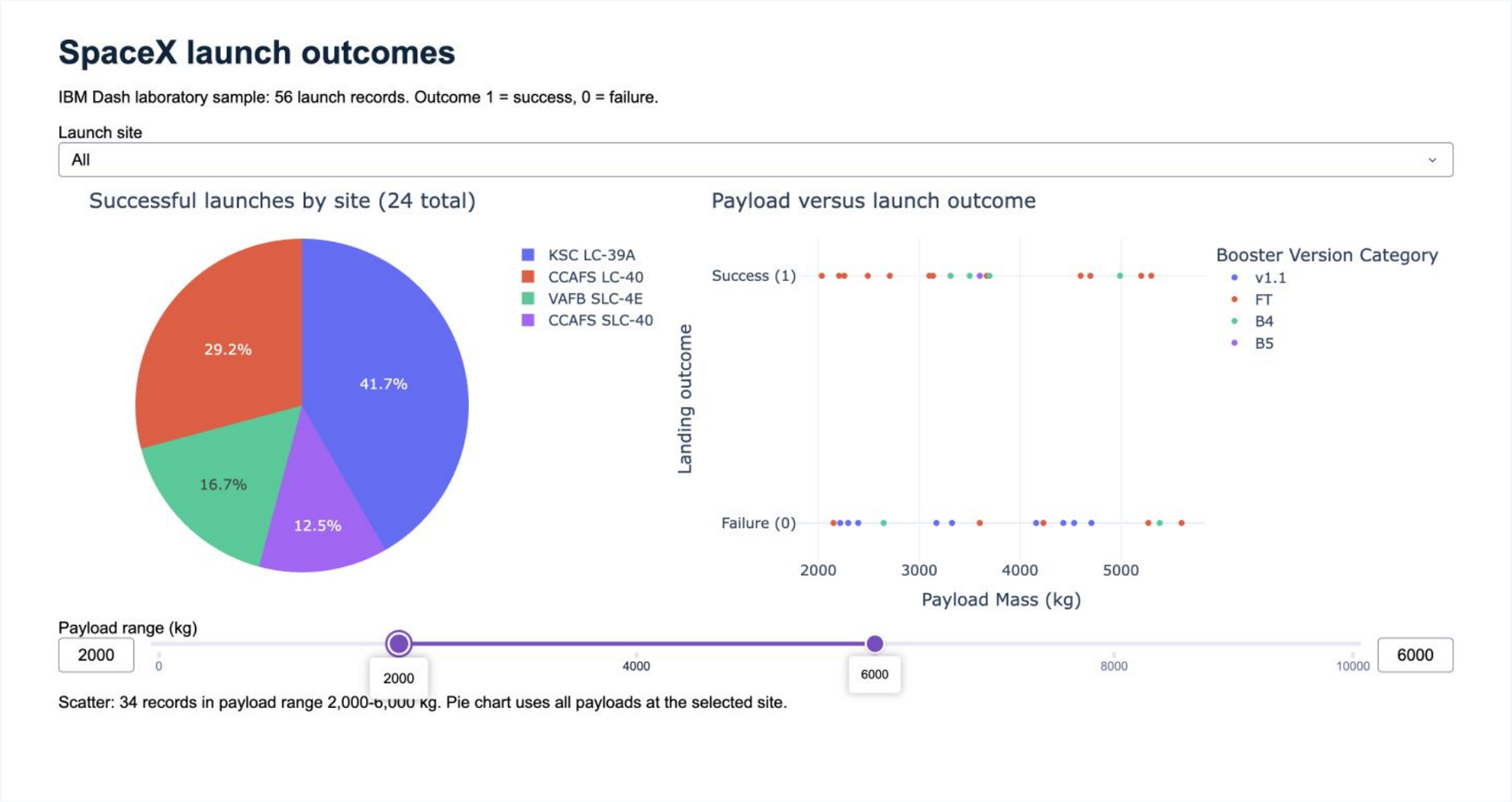
Pie chart for site with highest success ratio



KSC is selected in dropdown: 10 successes / 13 = 76.9%, highest site ratio in the 56-row dashboard sample.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/06_Plotly_Dash_Dashboard.ipynb

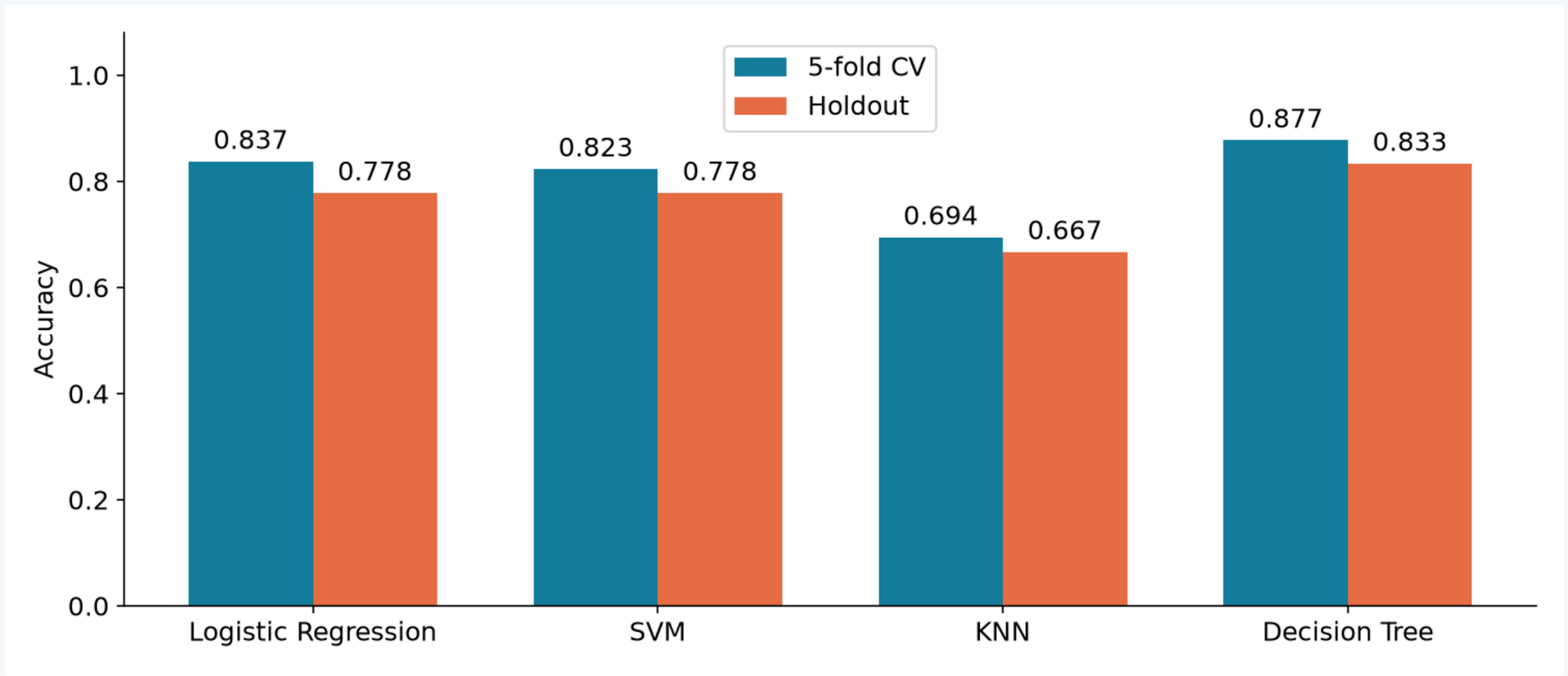
Payload vs launch outcome: two payload selections



All-site RangeSlider 0-10,000 kg: 56 records / 24 successes (42.9%; page 34). Range 2,000-6,000 kg: 34 records / 18 successes (52.9%). Colors identify booster version; this is descriptive.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/06_Plotly_Dash_Dashboard.ipynb

Classification Accuracy



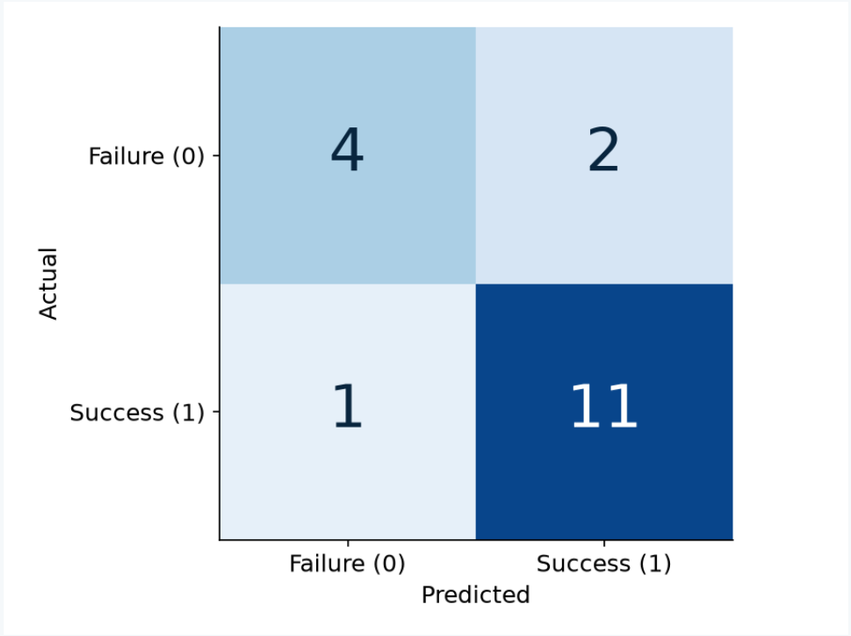
Model table reports five-fold CV accuracy and held-out test accuracy. Highest test accuracy: Decision Tree 0.833 (15/18).

Classifier	Accuracy	Precision	Recall	F1
Logistic Regression	0.778	0.786	0.917	0.846
SVM	0.778	0.786	0.917	0.846
KNN	0.667	0.667	1.000	0.800
Decision Tree	0.833	0.846	0.917	0.880

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/07_Modeling.ipynb

ACTUAL ROWS | PREDICTED COLUMNS | POSITIVE CLASS = SUCCESSFUL LANDING

Confusion Matrix: selected Decision Tree



[[TN=4, FP=2], [FN=1, TP=11]]. Correctly predicted 4/6 unsuccessful landing outcomes and 11/12 successful outcomes.

Precision=11/13=84.6%; recall=11/12=91.7%; F1=0.880; accuracy=15/18=83.3%. The small 18-launch holdout makes these estimates uncertain.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/07_Modeling.ipynb

Conclusion: findings, implications and next steps

• The Decision Tree outperforms the 66.7% majority baseline on this random holdout by three predictions, but two of six failures are missed.

• Modeling sample: 60/90 successful landings (66.7%). Dashboard sample: 24/56 (42.9%). Different coverage changes the baseline; samples are not pooled.

• Within the dashboard sample KSC has the highest site rate (10/13). This does not isolate site effect from flight time, payload and mission mix.

• Filtering the dashboard to 2,000-6,000 kg yields 18/34 successes (52.9%) versus 24/56 (42.9%) overall. The association does not show a causal payload effect.

• Limitations: small holdout, correlated flight/hardware/time, uneven categories and random split. Next: chronological evaluation, pre-launch features only, calibration and larger independent data.

https://github.com/joaoltrevisan/spacex-falcon9-capstone/blob/main/notebooks/07_Modeling.ipynb

Appendix: insight, sources and completed notebooks

Original insights: the success-rate baseline differs by 23.8 percentage points between the 90-row modeling sample and 56-row dashboard sample; the selected payload interval changes dashboard rate by 10.0 percentage points. Both illustrate how denominators and filters shape conclusions.

Repository

https://github.com/joaoltrevisan/spacex-falcon9-capstone

notebooks/01_Data_Collection_API.ipynb

notebooks/02_Data_Collection_Web_Scraping.ipynb

notebooks/03_Data_Wrangling_and_Target.ipynb

notebooks/04_EDA_and_Feature_Engineering.ipynb

notebooks/05_SQL_EDA_and_Folium_Map.ipynb

notebooks/06_Plotly_Dash_Dashboard.ipynb

notebooks/07_Modeling.ipynb

SpaceX API v4 docs: https://docs.spacexdata.com/ | Report builder and source code are in repository src/.